

VATREEL · DESIGN DECISIONS REGISTER

VatReel design decisions register

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	VTR-DEC-001	Version	0.1
Status	Draft	Date	2026-10-03
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/vatreel		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened; every decision made under Amish's 2026-10-03 pre-approval

VatReel design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

■ Open decisions

None. All decisions were made under Amish's 2026-10-03 pre-approval.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The stainless 17 and 51 tooth sprockets come bored 40 and 48.3 with hubs long enough for a roll pin	Sets the jackshaft and axle drilling and the guard case clearance	VTR-DDR-002, change 2
2	Pitting of the 304 chain 2 and 316L coupons in the partner's pickling and chrome liquors	R6 is at risk until shown; chain 2 may move to a 316 chain or a polymer drive	VTR-CAL-001, section 11
3	The bronze Tr24 x 5 nut bought is single start and runs self-locking on the rod under load	R11 depends on it; a two-start nut would not lock	VTR-CAL-001, section 6
4	Running clearance and creep of the glass-filled PTFE axle bushes in 98 °C water	R7; sets the bush bore	VTR-CAL-001, section 11
5	Temperature at the UHMW-PE bushes above a hot bath	UHMW-PE is good to about 80 °C; PTFE bushes there if it runs hotter	VTR-CAL-001, section 11
6	The first partner's pit: rim strength, width, length (at least 1.9 m), liquor level and end wall	The pivot sits 350 in from the end wall; the stands bear on the rims	VTR-REQ-001, assumptions
7	How the partner's workers hang hides (toggles through a punched hole) and loop yarn hanks on the bars	Sets the eye tab spacing and the snap hook size	VTR-DDR-001, item 7

■ Value engineering

Value-engineering target: USD 1,500 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 1,437 (USD 63 under the target). Main cost drivers and savings worth trying:

- Stainless is most of the cost: the reel and carrier bars (USD 246), the lift screw set (USD 130) and the bearing tower (USD 125). The frame (base, tower, shield, far stand, tubes) is 304; only parts that enter the liquor are 316L.
- Bought drive parts (sprockets USD 97, chains USD 62, bushes USD 63) are about 15 % of the cost; the stainless pair on the arm costs twice the plated pair in the tower.
- Making the design constructable added about USD 40: the closed chain 2 case, the ratchet cover and the wider splash shield. They are guarding, so savings are sought elsewhere.
- Savings worth trying: a hot-dip galvanized mild steel near stand and far stand where only splash reaches (about USD 120 less, to be weighed against corrosion in lime and acid splash); a bought cast iron handwheel in place of a stainless one; carrier bar end plates cut from offcuts.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	Reel on two arms swinging on a static telescopic pivot tube across the pit, not on a turning shaft between end stands	Amish, pre-approval: "I pre-approve the batch runs along with any recommendations you come up with."	VTR-DDR-001, item 1
2026-10-03	Reel axis across the pit, pointing at the operator; reel 1.18 m over the spokes, 0.6 m long; pits at least 1.9 m long	Amish, pre-approval (as above)	VTR-DDR-001, item 2
2026-10-03	Self-locking lift screw with a handwheel in place of a free lift lever (safety, conservative)	Amish, pre-approval (as above)	VTR-DDR-001, item 3
2026-10-03	Two-stage roller chain drive, 6 to 1, crank fixed 1.0 m above the floor	Amish, pre-approval (as above)	VTR-DDR-001, item 4
2026-10-03	Ratchet and pawl on the crank shaft as the holding brake	Amish, pre-approval (as above)	VTR-DDR-001, item 5
2026-10-03	304 frame, 316L wetted parts, PTFE bushes in the liquor, UHMW-PE above it	Amish, pre-approval (as above)	VTR-DDR-001, item 6
2026-10-03	Carrier bars with eye tabs and stainless snap hooks; yarn looped over the bars	Amish, pre-approval (as above)	VTR-DDR-001, item 7
2026-10-03	Enclose both chains, the sprockets and the ratchet; add R12 (safety, conservative)	Amish, pre-approval (as above)	VTR-DDR-001, item 8
2026-10-03	Two people present at any bath above 60 °C (safety, conservative)	Amish, pre-approval (as above)	VTR-DDR-001, item 9
2026-10-03	Own chain drive rather than the hand capstan block; CalRig the first candidate for the proof load	Amish, pre-approval (as above)	VTR-DDR-001, item 10
2026-10-03	R11 (lift) and R12 (guarding) added; pit length added to the assumptions	Amish, pre-approval (as above)	VTR-DDR-001, item 11
2026-10-03	First co-design candidate to approach: a worker health NGO or tannery association in the Kanpur leather cluster, India (not agreed)	Amish, pre-approval (as above)	VTR-DDR-001, item 12
2026-10-03	Value-engineering target kept at USD 1,500	Amish, pre-approval: "I also accept any cost overruns or variations from the assumed scope cost."	VTR-DDR-001, item 13
2026-10-03	Design for construction: torque tube, sprocket sizes, closed chain 2 case, ratchet cover, open far saddle with	Amish, pre-approval (as above)	

Date	Decision	Decided by	Record
	keeper, plug on the jackshaft tip, extension tube and clamp, wider shield and the other changes in VTR-DDR-002		VTR-DDR-002
2026-10-03	R8 accepted as not met for the prototype: two modules of 27.1 and 29.6 kg carried by two people (14.8 kg each at most); first setup about 29 minutes	Amish, pre-approval (as above)	VTR-CAL-001, section 9